TELESCOPE ASTRONOMY

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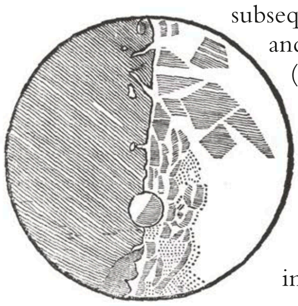
TELESCOPES ARE THE ULTIMATE optical instruments for astronomy. The simplest spyglass type has changed comparatively little over the centuries, but the most sophisticated amateur instruments now offer the optics and computerized controls once the preserve of professionals.

EARLY TELESCOPES

The invention of the telescope is usually credited to a Dutch optician named Hans Lippershey (1570–1619). In 1608, Lippershey found that a certain combination of optical lenses mounted at either end of a tube magnified an image—the basis of the refracting telescope. News of this device spread across Europe, and a year later the Italian scientist Galileo Galilei (1564–1642) built telescopes that could magnify up to 30 times. His subsequent observations of the Moon, Sun, and stars helped establish the heliocentric (Sun-centered) theory of the universe proposed by Copernicus (see p.69). In 1668, Isaac Newton developed the reflecting telescope, which used mirrors instead of lenses. There were many advantages: they did not have the optical defects that the refracting instruments had, tubes were shorter, and they could be made with larger apertures. However, the early mirrors were made of metal and tarnished, so they did not catch on initially.



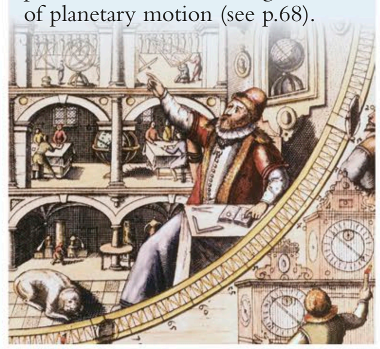
GALILEO'S SKETCH OF THE MOON
The Italian astronomer used his telescopes to observe the craters, mountains, and dark lowland areas of the Moon.

NEWTON'S TELESCOPE
Isaac Newton developed this first reflecting telescope, and his design is still used. A mirror at one end of the tube "bounces" light toward the eyepiece at the other end.

EXPLORING SPACE

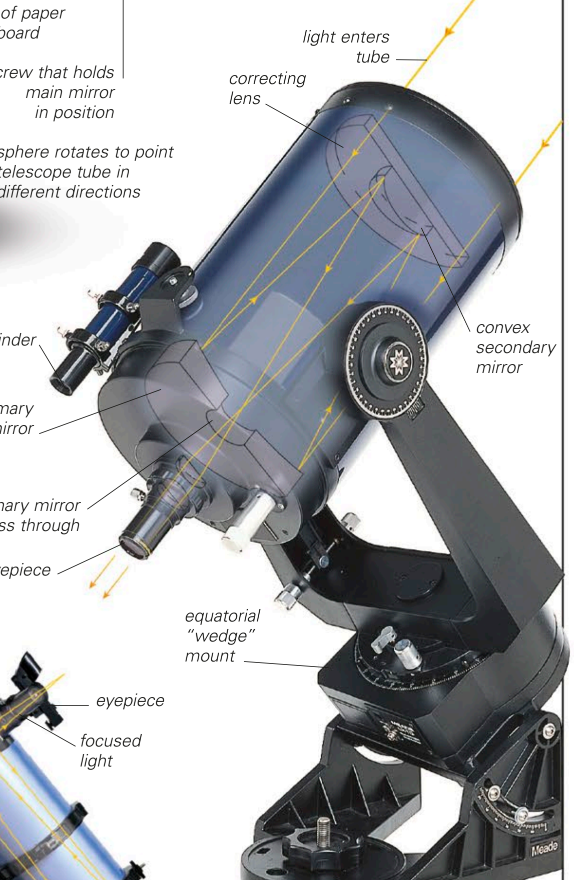
BEFORE THE TELESCOPE

In the days before telescopes, astronomers used a variety of instruments for measuring the positions of celestial objects. Tycho Brahe (1546–1601), a Danish nobleman, built his own observatory and equipped it with the finest instruments, which included a huge wall-mounted quadrant. German mathematician Johannes Kepler used Brahe's measurements of planetary positions when calculating his laws of planetary motion (see p.68).

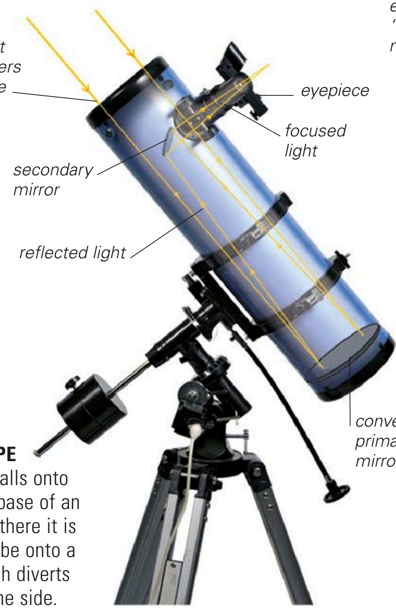


TELESCOPE DESIGNS

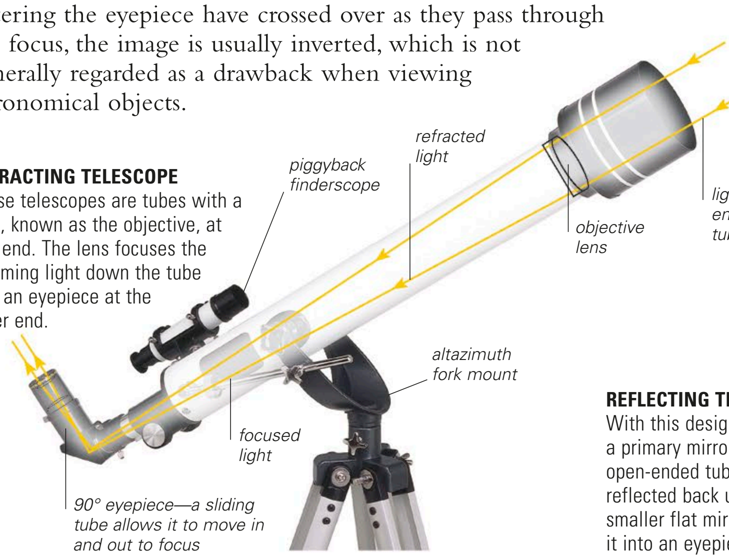
A telescope's function is to collect light from distant objects, bring it to a focus, and then magnify it. There are two basic ways of doing this, using either a lens or a concave mirror. A lens refracts, or bends, the light passing through it, directing it to a focal point behind it. A curved mirror reflects light rays back onto converging paths that come to a focus somewhere in front of it. A combination design called a catadioptric telescope is basically a reflector with a thin lens across the front of the tube. Light rays entering a telescope from astronomical objects are near parallel. Once the captured light rays have passed the focus, they begin to diverge again, at which point they are captured by an eyepiece, which returns the rays to parallel directions, magnifying them in the process. Because light rays entering the eyepiece have crossed over as they pass through the focus, the image is usually inverted, which is not generally regarded as a drawback when viewing astronomical objects.



CATADIOPTRIC TELESCOPE
In this compact reflector design, a convex secondary mirror directs light to the eyepiece through a hole in the primary mirror. By bouncing the light back on itself, the length of the telescope tube is reduced.



REFLECTING TELESCOPE
With this design, light falls onto a primary mirror at the base of an open-ended tube. From there it is reflected back up the tube onto a smaller flat mirror, which diverts it into an eyepiece on the side.



REFRACTING TELESCOPE
These telescopes are tubes with a lens, known as the objective, at one end. The lens focuses the incoming light down the tube into an eyepiece at the other end.